

ProGuard Obfuscation Bypass – Complete Guide

ProGuard ek popular obfuscation tool hai jo Android apps ke code ko **renaming obfuscation** ke zariye protect karta hai . Iska matlab hai ke class names, method names, aur field names ko short meaningless strings mein convert kar diya jata hai . Lekin yeh important hai ke ProGuard sirf renaming karta hai, actual opcodes (code logic) unchanged rehte hain . Isliye bypass possible hai.

ProGuard Obfuscation Kya Karti Hai?

Shrinking: Unused code ko remove karti hai, jo third-party libraries ke liye useful hai .

Obfuscation: Classes, fields, aur methods ko short meaningless names mein rename karti hai . Ye security ke liye nahi, balke minification (code size reduce) ke liye hai .

Optimization: Inlining, outlining, class merging jaise techniques se code ko aur shrink karti hai .

Mapping File: Obfuscation ke dauran ek mapping.txt file generate hoti hai jo original names aur obfuscated names ko link karti hai .

Bypass Method 1: ReTrace Tool (Stack Trace De-obfuscation)

ReTrace ProGuard ka companion tool hai jo obfuscated stack trace ko restore karta hai .

Usage:

```
java -jar retrace.jar mapping.txt stacktrace.txt
```

Options:

- -verbose – Method return types aur arguments ke saath detailed output .
- -regex – Custom regular expression specify karne ke liye .

Example Output (With Line Numbers):

Obfuscated Stack Trace:

```
java.io.IOException: Can't read [dummy.jar]  
    at proguard.y.a(MyApplication:188)  
    at proguard.y.a(MyApplication:158)
```

De-obfuscated Stack Trace:

```
java.io.IOException: Can't read [dummy.jar]
    at
proguard.InputReader.readInput(InputReader.java:18
8)
    at
proguard.InputReader.readInput(InputReader.java:15
8)
```

Without Line Numbers: Agar line numbers preserve nahi ki gayi hain toh ReTrace multiple possible original method names list karega . Aapko manually deduce karna hoga based on program logic.

Bypass Method 2: Custom Deobfuscation Scripts

Aapko ek mapping file chahiye. Agar available hai toh Python script se de-obfuscate kar sakte hain:

```
import re

def deobfuscate_line(line, mapping):
    for orig, obf in mapping.items():
        if obf in line:
            line = line.replace(obf, orig)
    return line

def load_mapping(mapping_file):
    mapping = {}
    with open(mapping_file, 'r') as f:
        for line in f:
```

```

        if '->' in line:
            parts = line.strip().split(' -> ')
            orig = parts[0].strip()
            obf = parts[1].strip()
            mapping[obf] = orig
    return mapping

# Usage
mapping = load_mapping('mapping.txt')
with open('obfuscated_code.txt', 'r') as f:
    for line in f:
        print(deobfuscate_line(line, mapping))

```

Bypass Method 3: JADX + Manual Analysis

1. APK ko decompile karein:

```
jadx-gui app.apk
```

2. Search functionality use karein (Ctrl+Shift+F):

- Interesting strings search karein (URLs, API keys, "password", "token")
- Call hierarchy search karein
- Class relationships understand karein

3. Manual mapping create karein:

- AndroidManifest.xml se entry point pata karein (MainActivity)
- JADX mein us class ke package name aur method signatures note karein

- Smali code mein obfuscated names search karein

Bypass Method 4: Advanced Obfuscation Bypass

dProtect-like Obfuscation: dProtect ProGuard ka fork hai jo advanced features provide karta hai: string obfuscation, constant obfuscation, arithmetic obfuscation, control flow obfuscation .

String Obfuscation Bypass: dProtect strings ko dynamic decryptor methods mein convert karta hai. JEB decompiler ki built-in optimizers automatically clean up kar deti hain .

Arithmetic Obfuscation Bypass: Constants ko XOR operations aur static array elements se replace kar diya jata hai. Custom IR plugins use karke static array elements ko actual values se replace kar sakte hain .

Constants Scrambling Bypass: Synthetic static arrays use hoti hain. Data flow analysis ke zariye variable references track karke static array elements identify aur replace kar sakte hain .

Important Tips

Mapping File Available Ho: Agar mapping file available hai toh de-obfuscation bohot easy hai. Agar nahi hai toh manual analysis ka sahara lena padta hai .

Line Numbers Preserve: Agar line numbers preserve ki gayi hain toh ReTrace fully accurate result deta hai. Agar nahi toh ambiguous ho jata hai .

ProGuard vs R8: R8 ProGuard ka re-implementation hai jo Android Studio mein default hai .

AI-Based Obfuscation: AI-based ProGuard obfuscation bhi use ho rahi hai, jahan meaningless Korean strings jaise obfuscation use hoti hain .

 **Educational Purpose Only:** Yeh guide sirf educational aur learning purposes ke liye hai. Obfuscation bypass ka istemal sirf apne own apps aur authorized environments mein karein. Kisi doosre ke app ko bina permission ke reverse engineer karna applicable cyber laws ke under punishable hai.

MADE BY:

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